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Algorithmic Intermediation and the International Transmission of U.S. Monetary Policy

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Abstract

This paper examines how algorithmic and AI-driven fund management shapes the international transmission of U.S. monetary policy to emerging markets. It argues that the key source of instability is not algorithmic intermediation itself, but the similarity of models across funds. When algorithms rely on similar signals and make correlated errors, their trades reinforce one another and intensify capital-flow responses during periods of stress. When models are diverse, errors offset each other and algorithmic investors can stabilize flows. The paper develops a two-region macro-financial framework and tests its central prediction using equity portfolio flows to nineteen emerging markets from 2000 to 2024. The evidence shows that algorithmic herding amplifies outflows after U.S. monetary shocks only in high-volatility regimes, while faster adjustment alone has no comparable effect. The amplification is negative in sign across an extensive robustness battery and statistically significant under the baseline specification, with two qualifications we state openly: significance depends on defining the stress regime by the VIX rather than by calendar years, and it weakens when the 2008–09 crisis—the most informative stress episode—is excluded. The results imply that policy should focus on preserving model diversity rather than limiting the size of non-bank intermediation.

Keywords: algorithmic intermediation; non-bank financial intermediaries; global financial cycle; U.S. monetary policy transmission; herding; capital flows; emerging markets; DSGE.

JEL classification: F32; E52; F42; G23; G15.

1 Introduction

This paper asks whether algorithmic and AI-driven fund management changes the international transmission of U.S. monetary shocks. The question matters for international financial stability, and the answer is not obvious, because it depends on how AI shapes financial decisions. On the one hand, common AI models can raise market correlations and amplify stress episodes, and correlated positioning already appears among the vulnerabilities that supervisors flag as a source of systemic risk (Financial Stability Board, 2024a). On the other, adaptive strategies that run on genuinely different architectures and data have been shown to supply liquidity when conventional investors flee, stabilizing flows rather than amplifying them. The same technology, then, looks like a source of fragility in some hands and of resilience in others.

The answer lies not in the presence of algorithms but in their *similarity*. When many funds run correlated models—trained on similar data, reading the same signals through the same pipelines—their forecast errors move together, and their trades compound. When funds run diverse models, their errors cancel, and the sector as a whole prices fundamentals more accurately than any single participant. A single parameter, which we call *algorithmic homogeneity*, captures this distinction: it is the fraction of forecast-error variance that algorithmic funds share.

Three facts about the current environment motivate both the modeling choice and the empirical work. First, stress episodes in cross-border capital flows are increasingly synchronized and fast: a small common trigger sets off a violent, correlated response across otherwise unrelated markets, driven by joint deleveraging rather than by country fundamentals. Second, the intermediaries capable of synchronized, model-driven trading are no longer marginal. According to the Financial Stability Board, the non-bank financial intermediation (NBFI) sector reached 49.1 percent of total global financial assets in 2023, having grown 8.5 percent that year—more than double the pace of the banking sector (Financial Stability Board, 2024b). By 2024 the share had risen to 51.0 percent, or 256.8 trillion dollars (Financial Stability Board, 2025). The fastest-growing component is the group that houses hedge funds, money market funds, and other investment vehicles—precisely the entities most likely to run algorithmic strategies. Third, the sensitivity of emerging-market capital flows to the global factor is not constant: fundamentals dominate in tranquil times, but in stress episodes the global factor becomes the dominant driver.

This paper develops a two-region DSGE model and derives three results. First, homogeneity amplifies the transmission of U.S. monetary shocks, but through the *variance* of the aggregate lending response rather than its mean: correlated errors survive aggregation while independent errors wash out. Second, heterogeneity attenuates transmission, and in the limit of fully diverse models the algorithmic sector becomes a perfectly informed, stabilizing intermediary. Third, and most important for what follows, the threshold level of homogeneity at which amplification sets in is *state-dependent*: it falls when local fundamentals deteriorate. This delivers a sharp and testable prediction. The same algorithmic sector that calms capital flows in normal times amplifies them in crises—not because homogeneity itself changes, but because the threshold it must cross drops when

the economy is under stress.

We test this prediction on a panel of equity portfolio flows to nineteen emerging markets over 2000–2024. Using a triple-interaction design that lets the effect of U.S. monetary shocks vary with algorithmic behavior and with the global stress regime, we find that the herding dimension of algorithmic behavior amplifies capital outflows, and that this amplification is confined to high-volatility periods, exactly as the state-dependent threshold predicts. The negative sign of the amplification is present in every specification of an extensive robustness battery, and it is statistically significant under the baseline specification and under the inference schemes best suited to the setting. We are equally explicit about the limits of that battery: significance rests on two design choices we document openly—defining the stress regime by the VIX rather than by calendar years, and retaining the 2008–09 crisis, the single most informative stress episode. A state-dependent mechanism that draws its identification from a small number of stress quarters should be expected to lose precision precisely when those quarters are redefined or removed, and we report those sensitivities rather than obscure them.

The contribution is threefold, and it is worth stating precisely how it differs from a fast-growing parallel literature on algorithmic homogeneity. Relative to the literature on the Global Financial Cycle (Rey, 2015; Miranda-Agrippino & Rey, 2020), which documents that U.S. monetary policy drives a common factor in global asset prices and flows, we provide a micro-founded mechanism—the aggregation of correlated forecast errors—that explains why the strength of transmission varies over time and across regimes. Relative to the literature on non-bank financial intermediation, which has documented the sector's growth and its procyclicality, we show that what matters for amplification is not the institutional form of the intermediary but the behavioral correlation across intermediaries. And relative to the recent structural work on algorithmic monoculture (Meng & Chen, 2026; Qiu & Han, 2026)—which models the correlation of machine behavior as a driver of aggregate systemic risk and validates it on U.S. institutional holdings—our distinct object is the *international transmission* of monetary policy: we embed the correlation of forecast errors in a two-region model of the U.S.–periphery cycle, an environment those frameworks do not address. The sharpest point of departure is qualitative. That literature derives an effect of algorithmic adoption on risk that is monotone and increasing—more homogeneity, more fragility. Our central result is instead a *state-dependent sign reversal*: the same algorithmic sector stabilizes capital flows when fundamentals are healthy and amplifies them only once fundamentals deteriorate and the amplification threshold falls below prevailing homogeneity. It is this conditional reversal, not the level effect of homogeneity, that we take to the data.

The remainder of the paper is organized as follows. Section 2 situates the contribution in the literature. Section 3 develops the two-region DSGE model, and Section 4 derives the three propositions. Section 5 establishes the robustness of the analytical results, and Section 6 reports the calibration. Section 7 tests the model's central prediction on a panel of equity portfolio flows to nineteen emerging markets. Section 8 draws out the implications for macroprudential policy and concludes.

2 Related Literature

Our contribution connects to three strands of literature.

The first is the literature on the Global Financial Cycle. Rey (2015) established that a common factor drives risky asset prices, capital flows, and leverage worldwide, comoving with U.S. monetary policy and the VIX, and Miranda-Agrippino & Rey (2020) showed through high-frequency identification that U.S. monetary contractions tighten financial conditions abroad even under floating exchange rates. This work motivates our center–periphery structure and the state-dependence of Fact 3, but it treats the intermediation sector as a black box. Our contribution is to open it: we show that the *composition* of intermediation—specifically the behavioral correlation across algorithmic funds—determines whether a given U.S. shock is amplified or dampened, providing a micro-founded reason why transmission strength varies over time.

The second is the literature on heterogeneous financial intermediaries. Coimbra & Rey (2024) develop a macro-finance model in which the cross-section of Value-at-Risk constraints generates a non-monotonic relationship between interest rates and financial stability, with heterogeneity in *risk-taking capacity* as the driving force. Our model shares the emphasis on cross-sectional heterogeneity but locates it elsewhere: not in how much risk intermediaries are willing to bear, but in how correlated their signal-extraction errors are. This distinction matters because our source of heterogeneity—model homogeneity ϕ—can be high even when risk appetites are identical, and it is ϕ, not risk capacity, that our propositions show to govern amplification. The broader intermediary-based macro-finance tradition provides the natural point of comparison for our state-dependent result. Brunnermeier & Sannikov (2014) show that an economy with a constrained financial sector exhibits highly non-linear amplification and occasional volatile crises—their "volatility paradox"—while He & Krishnamurthy (2013) model risk premia that rise non-linearly when intermediary equity capital becomes scarce. In both, the non-linearity is driven by the tightness of a balance-sheet or capital constraint. Our Proposition 3 delivers an analogous state-dependent non-linearity from an entirely different primitive: not the scarcity of capital, but the correlation of forecast errors across intermediaries. The two mechanisms are complementary, and in a fuller model a binding capital constraint and rising model homogeneity would reinforce one another.

The third is the literature on information choice and attention. Kacperczyk et al. (2016) provide a rational theory in which fund managers optimally choose which signals to learn about, tying information choices to the business cycle. We build on their information-theoretic foundation but invert one assumption: in our setting funds observe the *same* signals and differ in the correlation of their noise. This is the formal device—anchored in the exact law of large numbers for a continuum (Sun, 2006)—through which common forecast errors survive aggregation and produce herding. The state-dependence of signal precision that drives our threshold result follows the mechanism of Orlik & Veldkamp (2024), in which real-time estimation of non-normal distributions makes perceived uncertainty rise as outcomes move away from the center.

Two further connections position the contribution. The concern that common AI models raise

market correlations and amplify stress is now explicit in official assessments (Financial Stability Board, 2024a), but that literature is largely descriptive; we supply the analytical mechanism behind the concern. And the amplification of correlated flows in inelastic markets, quantified by Gabaix & Koijen (2021), is complementary to our channel: their price multiplier operates on the demand side of asset markets, while ours operates on the correlation of intermediary lending, and the two would compound in a fuller model. Finally, our mechanism is distinct from the one in the emerging literature on strategic AI behavior. Dou et al. (2025) show that reinforcement-learning trading algorithms can learn to collude—to sustain coordinated, supra-competitive behavior without explicit communication. Their coordination is strategic and emerges from repeated interaction; ours is statistical and emerges from shared training, requiring no strategic intent. The two are complementary faces of the same underlying concern, homogeneous machines behaving alike, and our contribution is to provide the DSGE micro-foundation for the non-strategic, correlation-of-errors channel that a general-equilibrium transmission model requires.

Closest to our object of study is a recent pair of structural papers on algorithmic monoculture, and it is important to be explicit about where we overlap and where we depart. Meng & Chen (2026) embed correlated algorithmic signals in a Kyle-type market with performative feedback and endogenous adoption, deriving a systemic-risk multiplier that grows superlinearly in the AI-adoption share and validating it on the universe of SEC Form 13F holdings with a shift-share instrument. Qiu & Han (2026) construct a multi-agent model whose AI traders share a representation layer, and argue that *representation* homogeneity—similarity in how agents encode market states—compresses the space of forecast disagreement under stress even when realized predictions look diverse in calm times. Our forecast-error homogeneity ϕ is, in their taxonomy, the *forecast-overlap* object rather than the representation object; we take as our primitive precisely the quantity their framework treats as secondary, but we do so in a setting neither addresses. Three differences are substantive rather than terminological. First, both model a single asset traded through a market maker; neither contains a central bank, a monetary shock, or a center–periphery structure, so the international transmission question we pose is outside their scope. Second, where they posit signal correlation as an exogenous parameter, we *derive* the survival of common noise under aggregation from the exact law of large numbers for a continuum (Sun, 2006), so that homogeneity operates through aggregation alone and not through how funds weight their signals. Third, and most important, their central prediction is monotone—adoption raises risk—whereas Proposition 3 delivers a reversal of sign across regimes. This last difference also frames a limitation we return to in Section 7: because their prediction is about tail risk, both papers test a second moment directly, while our panel evidence identifies a conditional-mean effect; aligning the two is the natural next step for the empirical design rather than a settled matter.

3 A Two-Region Economy with Two Types of Intermediary

The model has a center—the United States—that sets monetary policy, and a periphery—a representative emerging market—that receives cross-border lending. The single novel ingredient is the intermediation sector: alongside traditional funds that adjust lending sluggishly, a continuum of algorithmic funds extracts noisy signals about U.S. policy and local fundamentals and lends on the basis of those signals. A single parameter, the algorithmic homogeneity $\phi \in [0, 1]$, governs how correlated the funds' forecast errors are. The paper's central result is that this one parameter determines whether algorithmic intermediation amplifies or dampens the transmission of U.S. monetary shocks to the periphery.

Throughout, we reserve ϕ for algorithmic homogeneity alone. The inverse Frisch elasticity is denoted ν, and the Taylor-rule coefficients are φ_π and φ_y, so that each symbol carries a single meaning.

3.1 Households

A representative household in the periphery chooses consumption C_t and labor N_t to maximize

$$\mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \left[\frac{C_t^{1-\sigma}}{1-\sigma} - \chi \frac{N_t^{1+\nu}}{1+\nu} \right], \tag{1}$$

where $\beta \in (0, 1)$ is the discount factor, $\sigma > 0$ is the coefficient of relative risk aversion (the inverse of the intertemporal elasticity of substitution), $\nu > 0$ is the inverse Frisch elasticity of labor supply, and $\chi > 0$ scales the disutility of work. Two features of (1) matter for the transmission mechanism. First, σ controls how strongly the household smooths consumption across periods: a higher σ makes the household more reluctant to let consumption fluctuate, which flattens its response to the interest-rate movements that a U.S. shock induces, and therefore dampens the real-side amplification of any given financial shock. Second, ν governs how elastically labor responds to wage changes: a low ν (elastic labor) lets the periphery absorb shocks through hours worked rather than through consumption, again muting the propagation. These two curvature parameters are the household-side channels through which the financial disturbance at the center of the paper reaches the real economy. The household faces the budget constraint

$$C_t + D_t + I_t = W_t N_t + R_{t-1} D_{t-1} + R_t^K K_{t-1} + \Pi_t - \frac{\psi_I}{2} \left(\frac{I_t}{K_{t-1}} - \delta \right)^2 K_{t-1}, \tag{2}$$

and the capital law of motion $K_t = (1 - \delta) K_{t-1} + I_t$. Here D_t are deposits paying the gross rate R_t, I_t is investment, W_t is the real wage, R_t^K is the rental rate of capital, Π_t are profits rebated to the household, $\delta \in (0, 1)$ is the depreciation rate, and $\psi_I > 0$ governs the quadratic cost of adjusting investment. This adjustment cost prevents investment from jumping instantaneously to its new optimum when a shock hits, and thereby generates the persistent, hump-shaped investment response that the data display. Without it, the capital stock would absorb shocks too quickly and

the financial channel would leave no lasting imprint on the real economy; ψ_I is the parameter that ties the speed of real adjustment to the financial disturbance. The optimality conditions (derived step by step in the appendix) deliver the consumption Euler equation, the labor-supply schedule, and the investment Euler equation:

$$1 = \beta\, \mathbb{E}_t \left[\left(\frac{C_{t+1}}{C_t} \right)^{-\sigma} R_t \right], \tag{3}$$

$$\chi N_t^{\nu} C_t^{\sigma} = W_t, \tag{4}$$

$$Q_t = \beta\, \mathbb{E}_t \left[\left(\frac{C_{t+1}}{C_t} \right)^{-\sigma} \left(R_{t+1}^K + Q_{t+1}(1-\delta) \right) \right], \tag{5}$$

where Q_t is Tobin's marginal q, the shadow value of installed capital. Equation (3) is standard: the household equates the marginal utility cost of saving one unit today to the discounted marginal utility gain of consuming its gross return tomorrow—this is the channel through which the policy rate R_t, moved by the U.S. shock, reshapes the periphery's saving and consumption path. Equation (4) equates the marginal rate of substitution between leisure and consumption to the real wage, closing the labor-supply block. Equation (5) prices installed capital: the household holds capital up to the point where its shadow value Q_t equals the discounted sum of next period's rental return and its resale value net of depreciation. When Q_t rises—because expected returns to capital improve or the discount factor falls—investment expands, and it is through this valuation equation that the credit conditions set by the intermediation sector feed into real capital formation.

3.2 Firms

A representative firm produces output with the constant-returns technology

$$Y_t = A_t K_{t-1}^{\alpha} N_t^{1-\alpha}, \tag{6}$$

where $\alpha \in (0,1)$ is the capital share and total factor productivity A_t follows an AR(1) process, $\log A_t = \rho_A \log A_{t-1} + \varepsilon_t^A$. The persistence ρ_A determines how long a productivity disturbance lingers in the periphery's fundamentals, and it therefore interacts directly with the state-dependent signal precision of Section 3: the further and longer A_t (and with it the fundamental F_t) strays from steady state, the noisier local signals become and the lower the amplification threshold falls. Cost minimization yields the standard factor prices $W_t = (1-\alpha) Y_t / N_t$ and $R_t^K = \alpha Y_t / K_{t-1}$. Firms finance capital by borrowing from intermediaries at the loan rate $\tilde{R}_t^K = R_t + \mu_t$, where the external finance premium

$$\mu_t = \bar{\mu} \left(\frac{L_t}{\bar{L}} \right)^{\xi}, \qquad \xi > 0, \tag{7}$$

rises with aggregate lending L_t relative to its steady-state level $\bar{L}$. Equation (7) is the pivotal link between the financial sector and the real economy, and ξ is arguably the single most consequential parameter in the model. It measures how sharply the cost of external finance responds to the volume

of credit: when algorithmic funds withdraw lending in unison, L_t falls, the premium μ_t rises, and firms face a higher loan rate $\tilde{R}_t^K$ that chokes off investment. A large ξ means the premium is highly sensitive to credit volume, so a synchronized algorithmic retreat produces a violent tightening of financial conditions; a small ξ cushions the same retreat. This is the transmission belt through which the correlation of forecast errors, an object internal to the intermediation sector, becomes a macroeconomic contraction in the periphery.

3.3 Traditional funds

Traditional funds are slow. They choose lending L_t^T to maximize expected profits net of a quadratic cost of adjusting the loan book:

$$\max_{L_t^T} \; \mathbb{E}_t \left[\tilde{R}_t^K L_t^T - R_t L_t^T - \frac{\psi_T}{2} \left(\frac{L_t^T}{L_{t-1}^T} - 1 \right)^2 L_{t-1}^T \right]. \tag{8}$$

The first-order condition,

$$\mu_t = \psi_T \left(\frac{L_t^T}{L_{t-1}^T} - 1 \right), \tag{9}$$

makes traditional lending a smooth, backward-looking function of the premium: the loan book expands only gradually, and the adjustment cost ψ_T sets the pace. This sluggishness is the foil against which algorithmic speed is defined.

3.4 Algorithmic funds

A continuum of algorithmic funds indexed by $j \in [0, 1]$ observes noisy signals about the U.S. monetary policy shock and about local fundamentals. Fund j's signals are

$$s_{j,t}^{MP} = MP_t + \nu_{j,t}^{MP}, \tag{10}$$

$$s_{j,t}^{LOC} = F_t + \varepsilon_{j,t}^{LOC}, \tag{11}$$

where MP_t is the policy shock, F_t is the (economy-wide) periphery fundamental, and the noise terms decompose into a common component and a fund-specific idiosyncratic component:

$$\nu_{j,t}^{MP} = \sqrt{\phi}\, \zeta_t^{MP} + \sqrt{1-\phi}\, u_{j,t}^{MP}, \qquad \varepsilon_{j,t}^{LOC} = \sqrt{\phi}\, \zeta_t^{LOC} + \sqrt{1-\phi}\, u_{j,t}^{LOC}. \tag{12}$$

Here $\zeta_t^{MP}, \zeta_t^{LOC}$ are common shocks shared by all funds—the signature of models trained on the same data with the same architecture—while $u_{j,t}^{MP}, u_{j,t}^{LOC}$ are independent across funds.

Assumption 1 (Noise normalization)**.** *The common and idiosyncratic noise components have equal variance:* $\mathrm{Var}(\zeta_t^k) = \mathrm{Var}(u_{j,t}^k) = \sigma_k^2$ *for* $k \in \{MP, LOC\}$.

Assumption 1 carries analytical weight. Under (12), the variance of an individual fund's noise is $\mathrm{Var}(\nu_{j,t}^{MP}) = \phi\, \sigma_{MP}^2 + (1-\phi)\, \sigma_{MP}^2$, and the fraction of that variance attributable to the common component equals ϕ *only* when the two variances coincide. Without Assumption 1, the fraction is

$\phi\sigma_\zeta^2/[\phi\sigma_\zeta^2 + (1-\phi)\sigma_u^2] \neq \phi$, and the interpretation of ϕ in Definition 1 fails. We therefore impose it explicitly.

Definition 1 (Algorithmic homogeneity)**.** *The parameter $\phi \in [0,1]$ is the fraction of forecast-error variance that is common across algorithmic funds. When $\phi = 0$, funds produce fully independent errors (maximum diversity); when $\phi = 1$, all funds share the same error (maximum homogeneity).*

Each fund forms a minimum–mean-squared-error forecast of next-period periphery output, weighting each signal by its Bayesian precision:

$$\mathbb{E}^A_{j,t}[Y_{t+1}] = \theta^{MP}\, a\, s^{MP}_{j,t} + \theta^{LOC}\, b\, s^{LOC}_{j,t}, \tag{13}$$

where a, b are the loadings of output on the policy and fundamental states and the weights $\theta^{MP}, \theta^{LOC}$ are the signal-to-total precision ratios. Under Assumption 1 the variance of each fund's signal noise is σ_k^2, *independent of* ϕ. It follows that the Bayesian weights themselves do not depend on ϕ:

$$\frac{\partial \theta^{MP}}{\partial \phi} = \frac{\partial \theta^{LOC}}{\partial \phi} = 0. \tag{14}$$

This is a substantive result, verified symbolically, and it clarifies the mechanism: homogeneity does not operate by changing how funds weight their signals. It operates entirely through aggregation, as we now show.

The precision of the local signal is state-dependent. Following the mechanism in Orlik & Veldkamp (2024)—whereby real-time estimation of a distribution with non-normal tails makes perceived uncertainty rise as outcomes move away from the center—we let

$$\sigma^2_{LOC,t} = \sigma_0^2 + \kappa_F \left(F_t - \bar{F}\right)^2, \qquad \kappa_F > 0. \tag{15}$$

Local signals become noisier when fundamentals are far from their steady state, in either direction—deep recession or overheating. This single feature drives the regime dependence of Proposition 3.

Each fund chooses lending $L^A_{j,t}$ under a static mean–variance objective, yielding the linear rule

$$L^A_{j,t} = \bar{L}_A - \gamma_1 \hat{\mu}_t + \gamma_2\, \mathbb{E}^A_{j,t}[Y_{t+1}], \tag{16}$$

where $\hat{\mu}_t$ is the log-deviation of the premium, $\gamma_1 > 0$ is the sensitivity of lending to the premium, and $\gamma_2 > 0$ is the sensitivity to the output forecast. The choice of a static objective for algorithmic funds—in contrast to the intertemporal problem (8) of traditional funds—is deliberate and encodes the paper's central contrast: algorithmic funds react to current signals without internalizing adjustment costs, which is what makes them fast.

Aggregating (16) across j and applying the exact law of large numbers for a continuum (Sun,

2006), the idiosyncratic terms $u_{j,t}^{MP}, u_{j,t}^{LOC}$ integrate to zero, while the common terms survive:

$$L_t^A = \bar{L}_A - \gamma_1 \hat{\mu}_t + \gamma_2 \left[\theta^{MP} a \left(MP_t + \sqrt{\phi}\, \zeta_t^{MP} \right) + \theta^{LOC} b \left(F_t + \sqrt{\phi}\, \zeta_t^{LOC} \right) \right]. \tag{17}$$

Equation (17) is the heart of the model. The common noise $\sqrt{\phi}\,\zeta$ does not vanish under aggregation: it moves every fund in the same direction at once. This is the formal content of herding. When $\phi = 0$, the common terms disappear and the aggregate algorithmic forecast is exact—the funds behave as a single, perfectly informed, stabilizing intermediary.

3.5 Market clearing and monetary policy

Total lending is $L_t = L_t^T + L_t^A$, with households allocating a steady-state share ω of deposits to traditional funds and $1-\omega$ to algorithmic funds. (We treat ω as a fixed structural share; endogenizing the portfolio choice is a straightforward extension that does not alter the transmission results.) The U.S. central bank sets its policy rate according to an inertial Taylor rule:

$$R_t = \rho_R R_{t-1} + (1 - \rho_R) \left[\bar{R} + \varphi_\pi \pi_t + \varphi_y y_t \right] + \varepsilon_t^{MP}. \tag{18}$$

The rule combines two ideas. The bracketed term is the standard Taylor (1993) prescription: the central bank raises the nominal rate above its long-run level $\bar{R}$ whenever inflation π_t or the output gap y_t rises above target, with φ_π and φ_y measuring the strength of each response. The requirement $\varphi_\pi > 1$—the Taylor principle—ensures that the real rate rises when inflation does, which Woodford (2003) and Galí (2015) show is necessary for a determinate, stable equilibrium; we set $\varphi_\pi = 2.0$, consistent with the estimates of Clarida et al. (2000) for the post-Volcker period. The remaining term, $\rho_R R_{t-1}$, introduces interest-rate smoothing: the central bank moves the rate only partially toward its target level each period, adjusting gradually rather than in discrete jumps. This inertia is a robust empirical feature of policy and has an optimizing rationale—Woodford (2003) shows that history-dependent, inertial rules arise naturally when the central bank internalizes the effect of expected future rates on current conditions. The smoothing parameter $\rho_R \in [0, 1)$ governs the speed of adjustment; a higher ρ_R means a more gradual policy path. The term ε_t^{MP} is the monetary policy shock whose international transmission the paper studies: it is the innovation in the policy rate that is orthogonal to the systematic response to inflation and the output gap, and it is the disturbance that algorithmic funds in the periphery must read through their noisy signals.

4 Analytical Propositions

We now state the three propositions. Each is followed by a discussion. Proofs are collected in the appendix. The three results map directly onto the three stylized facts. Proposition 1 formalizes *Synchronization*: it identifies the survival of common forecast errors under aggregation as the mechanism that turns homogeneous machines into a source of amplified, correlated flows.

Proposition 2 shows the converse—that diversity across models is stabilizing—which is why *Scale* alone does not imply fragility: a large algorithmic sector is dangerous only when it is homogeneous. And Proposition 3 reproduces *State dependence*: the same sector attenuates transmission in tranquil times and amplifies it in crises, because the threshold separating the two regimes falls as fundamentals deteriorate.

Proposition 1 (Amplification under homogeneity)**.** *Consider the response of aggregate algorithmic lending to a one-standard- deviation contractionary monetary shock. The* mean *response is independent of ϕ, but the* variance *of the aggregate response is proportional to ϕ:*

$$\mathrm{Var}\left(\frac{\partial L_t^A}{\partial MP_t}\right) = \left(\gamma_2 \theta^{MP} a\right)^2 \phi\, \sigma_{MP}^2. \tag{19}$$

The amplification factor—the ratio of the standard deviation of the aggregate response under high homogeneity ϕ_H to that under low homogeneity ϕ_L—is

$$\mathcal{A}(\phi_H, \phi_L) = \sqrt{\phi_H / \phi_L}. \tag{20}$$

A monetary shock has a deterministic effect on the average fund's forecast, and that effect does not depend on how correlated the funds' errors are. What homogeneity changes is the *dispersion* of the aggregate response. When errors are independent they wash out and the aggregate is smooth; when they are correlated they compound, and the aggregate lending response inherits a common component that scales with $\sqrt{\phi}$. Amplification is therefore a statement about the volatility of capital flows, not about their average level—which is precisely why it shows up in stress episodes rather than in means. Evaluated at $\phi_H = 0.8$ and $\phi_L = 0.2$, the factor is exactly 2.0; across the plausible range it runs from about 1.4 to 3.0 (Table 2). Figure 1 plots the closed-form factor: it is a transparent function of the two homogeneity levels, rising with the stress homogeneity ϕ_H and falling with the normal-times homogeneity ϕ_L, and crossing unity—no amplification—only when the two coincide.

Proposition 2 (Attenuation under heterogeneity)**.** *The variance of the aggregate algorithmic forecast error is proportional to ϕ. As $\phi \to 0$, the aggregate forecast becomes unbiased with zero variance, and the herding channel vanishes:*

$$\lim_{\phi \to 0} \mathrm{Var}\left(\bar{\varepsilon}_t^A\right) = 0. \tag{21}$$

With diverse models, the mistakes of one fund are offset by the opposite mistakes of another. In the aggregate, the errors cancel and the algorithmic sector delivers an exact reading of fundamentals. Diverse algorithmic intermediation is thus not merely harmless—it is actively stabilizing, supplying countercyclical liquidity because it prices fundamentals correctly when correlated investors do not.

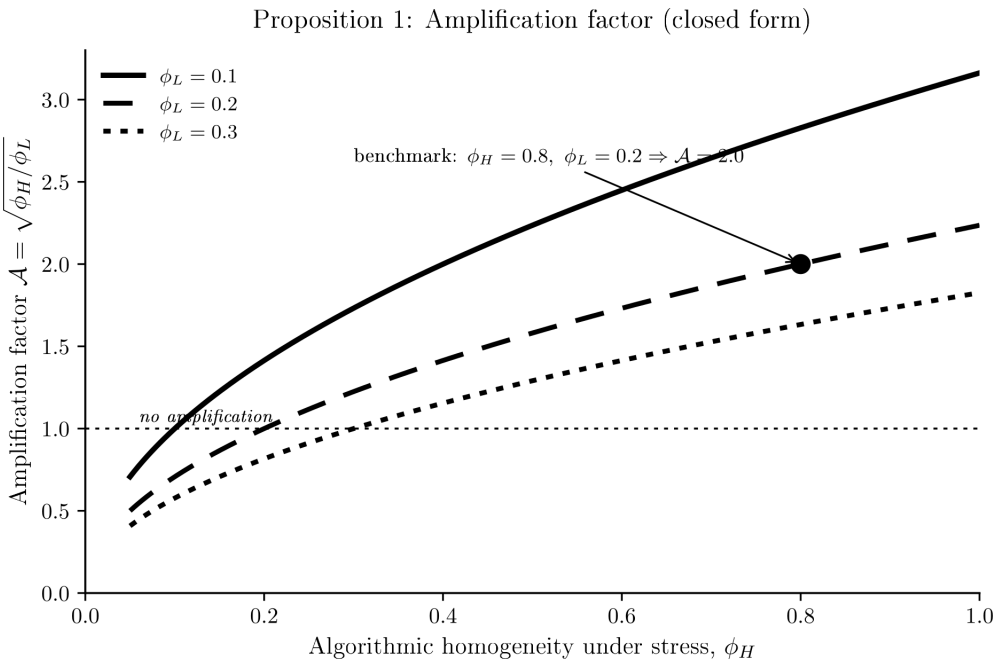


Figure 1: The amplification factor $\mathcal{A} = \sqrt{\phi_H/\phi_L}$ of Proposition 1, for three values of the normal-times homogeneity ϕ_L. The benchmark comparison ($\phi_H = 0.8$, $\phi_L = 0.2$) yields $\mathcal{A} = 2.0$.

Proposition 3 (State-dependent amplification threshold)**.** *Let the net effect of algorithmic intermediation on the comovement between capital flows and monetary shocks be the difference between an information-processing (attenuating) channel that scales with local signal precision $\tau_L = 1/\sigma^2_{LOC,t}$ and a herding (amplifying) channel that scales with ϕ. There is a threshold*

$$\phi^* = \frac{\mathcal{A}_{info}\,\tau_L}{\mathcal{A}_{herd}} = \frac{\mathcal{A}_{info}}{\mathcal{A}_{herd}\left(\sigma_0^2 + \kappa_F\,\Delta F^2\right)}, \qquad \Delta F \equiv |F_t - \bar{F}|, \tag{22}$$

such that the net effect is attenuating for $\phi < \phi^*$ and amplifying for $\phi > \phi^*$. The threshold falls as fundamentals deteriorate:

$$\frac{\partial \phi^*}{\partial \Delta F} = -\frac{2\,\mathcal{A}_{info}\,\kappa_F\,\Delta F}{\mathcal{A}_{herd}\left(\sigma_0^2 + \kappa_F \Delta F^2\right)^2} < 0. \tag{23}$$

The threshold in the previous proposition is the level of homogeneity at which the two channels exactly offset. In normal times, local signals are precise, the information channel is strong, and ϕ^* is high—so realistic levels of homogeneity leave transmission in the attenuating region. In a crisis, fundamentals move far from steady state, local signals lose precision, the information channel weakens, and ϕ^* collapses. Even moderate homogeneity now exceeds the threshold, and the same algorithmic sector that stabilized flows in calm times amplifies them under stress. This sign reversal—attenuation in normal periods, amplification in crises—is the model's sharpest prediction and the one the empirical section is designed to test. Note that the reversal is a property of the threshold crossing, not of any change in ϕ itself: homogeneity can be constant while its consequences

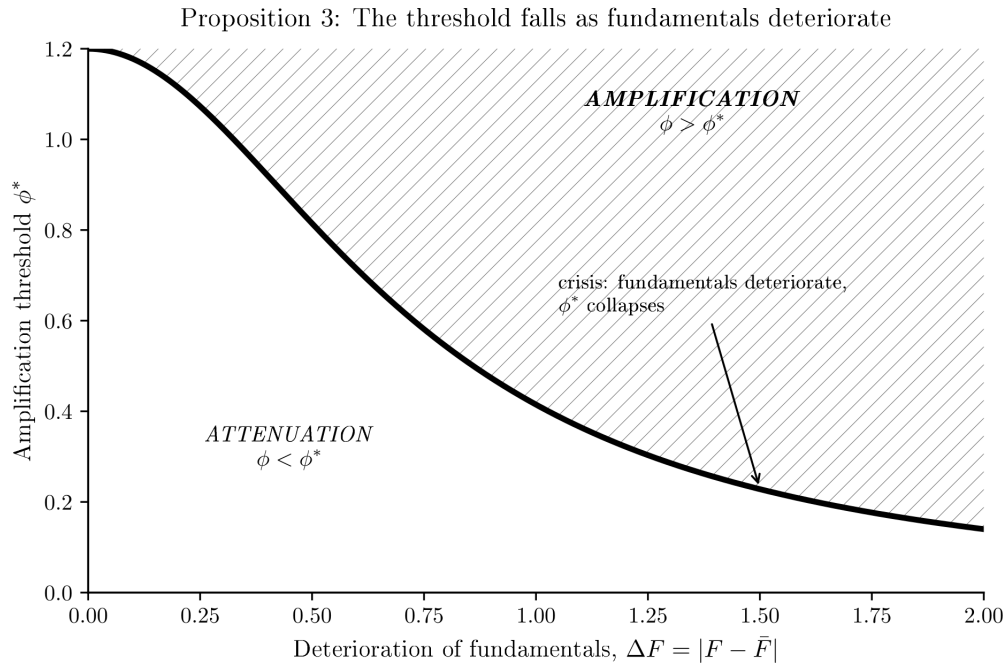


Figure 2: The state-dependent amplification threshold ϕ^* of Proposition 3. As fundamentals deteriorate (ΔF rises), ϕ^* falls, and a fixed homogeneity level crosses from the attenuation region into the amplification region.

flip with the state of the economy. Figure 2 makes the mechanism visible: the threshold ϕ^* traces a falling curve in the deterioration of fundamentals, dividing the $(\Delta F, \phi)$ plane into an attenuation region below and an amplification region above. A fixed level of homogeneity that sits in the attenuation region in calm times is overtaken by the collapsing threshold as fundamentals worsen.

5 Robustness of the Analytical Results

Three assumptions underlie the propositions: the equal-variance normalization of Assumption 1, the reduced-form spread (7), and the quadratic form of the state-dependent precision (15). We now show that none of the three central results—amplification, the amplification factor, and the falling threshold—depends on the specific form of these assumptions.

Consider first relaxing the equal-variance normalization. Assumption 1 sets $\sigma_\zeta^2 = \sigma_u^2$, which makes ϕ the exact fraction of common forecast-error variance. Suppose instead that the two variances differ, and let $\rho \equiv \sigma_\zeta^2/\sigma_u^2$ be their ratio. The variance of the aggregate common component becomes proportional to $\phi\rho$ rather than to ϕ, but the amplification factor—the ratio of the standard

deviation of the aggregate response under high versus low homogeneity—is

$$\mathcal{A} = \sqrt{\frac{\phi_H \, \rho}{\phi_L \, \rho}} = \sqrt{\frac{\phi_H}{\phi_L}}, \tag{24}$$

in which ρ cancels exactly. The amplification result of Proposition 1 is therefore invariant to the equal-variance normalization. What the normalization buys is interpretive convenience, not the result itself: without it, ϕ ceases to be literally the fraction of common variance and becomes a monotone proxy for homogeneity, while every comparative static goes through unchanged.

The real-side transmission runs through the external-finance premium (7), whose elasticity to credit volume is exactly ξ: differentiating, $\mathrm{d}\log\mu_t/\mathrm{d}\log L_t = \xi$. A synchronized algorithmic withdrawal of a given size therefore produces a premium increase that is strictly increasing in ξ, so the real amplification inherits the sign and monotonicity of the financial amplification for every admissible $\xi > 0$. The mechanism does not rely on a knife-edge value of the elasticity; it operates across its entire range, strengthening as the credit market becomes more inelastic—precisely the regime that the inelastic-markets literature identifies as empirically relevant.

Proposition 3 was derived under a quadratic precision loss, $\sigma^2_{LOC,t} = \sigma_0^2 + \kappa_F \Delta F^2$. Replace the quadratic with a general power, $\sigma^2_{LOC,t} = \sigma_0^2 + \kappa_F \Delta F^{\,p}$ for any $p > 0$. The threshold becomes $\phi^* = \mathcal{A}_{\text{info}}/[\mathcal{A}_{\text{herd}}(\sigma_0^2 + \kappa_F \Delta F^{\,p})]$, and its derivative,

$$\frac{\partial \phi^*}{\partial \Delta F} = -\frac{\mathcal{A}_{\text{info}} \, \kappa_F \, p \, \Delta F^{\,p-1}}{\mathcal{A}_{\text{herd}} \left(\sigma_0^2 + \kappa_F \Delta F^{\,p}\right)^2} < 0, \tag{25}$$

is negative for every $p > 0$ and every $\Delta F > 0$. The sign reversal that Proposition 3 predicts—attenuation in normal times, amplification in crises—is thus a property of the mechanism, not of the quadratic specification: any precision loss that is increasing in the distance of fundamentals from steady state delivers the same falling threshold. This is the sense in which the paper's sharpest prediction is structural rather than functional-form dependent.

6 Calibration

Table 1 reports the quarterly calibration. We distinguish sources carefully. Values labeled "standard" are conventional in the New Keynesian literature; values labeled "calibrated" are set to match a stated target and are not drawn from a specific external estimate; and values with a citation are taken from the cited source. Two parameter choices warrant explanation. First, the Taylor-rule inflation coefficient is set following Clarida et al. (2000), who estimate a forward-looking reaction function in which the response to expected inflation in the Volcker–Greenspan period is well above unity; we therefore use $\varphi_\pi = 2.0$ rather than the Taylor (1993) value of 1.5. Second, the state-dependence parameter κ_F is calibrated to the dispersion of emerging-market forecast errors; the mechanism is

motivated by Orlik & Veldkamp (2024), but the numerical value is not taken from that paper.

Table 1: Baseline calibration (quarterly)

Parameter	Value	Description	Source
β	0.99	Discount factor	Standard
σ	2.0	Risk aversion	Galí (2015)
ν	1.0	Inverse Frisch elasticity	Galí (2015)
δ	0.025	Depreciation rate	Standard
α	0.33	Capital share	U.S. national accounts
ψ_T	0.80	Traditional lending adjustment cost	Calibrated
γ_1	0.16	Algorithmic sensitivity to premium	Calibrated
γ_2	0.80	Algorithmic sensitivity to forecast	Calibrated
σ^2_{MP}	0.01	MP signal noise variance	HF surprise data
σ^2_0	0.05	Baseline local noise	EM forecast errors
κ_F	0.10	State-dependent noise curvature	Calibrated[a]
ϕ	$[0, 1]$	Algorithmic homogeneity	Varies
ω	0.70	Traditional-fund share	FSB NBFI data[b]
ξ	0.20	Spread elasticity	Credit-spread data
ρ_R	0.75	Interest-rate smoothing	Clarida et al. (2000)
φ_π	2.0	Taylor rule: inflation	Clarida et al. (2000)
φ_y	0.50	Taylor rule: output gap	Clarida et al. (2000)

Notes. [a] Mechanism motivated by Orlik & Veldkamp (2024); the value is calibrated to the dispersion of emerging-market forecast errors. [b] The traditional-fund share ω is set to the complement of the NBFI share of credit intermediation reported in the FSB Global Monitoring Report on Non-Bank Financial Intermediation.

The amplification factor of Proposition 1 depends only on the two homogeneity levels being compared, not on the rest of the calibration. Table 2 reports it across a grid. It exceeds three only at implausibly extreme homogeneity gaps ($\phi_H = 0.9$, $\phi_L \approx 0.1$); at the benchmark comparison ($\phi_H = 0.8$, $\phi_L = 0.2$) it equals 2.0. We report the factor as an analytical object rather than committing to a single number, since it is a transparent function of the homogeneity levels.

Table 2: Amplification factor $\mathcal{A} = \sqrt{\phi_H/\phi_L}$

	$\phi_L = 0.1$	$\phi_L = 0.2$	$\phi_L = 0.3$
$\phi_H = 0.6$	2.45	1.73	1.41
$\phi_H = 0.7$	2.65	1.87	1.53
$\phi_H = 0.8$	2.83	2.00	1.63
$\phi_H = 0.9$	3.00	2.12	1.73

7 Empirical Evidence

The model leaves us with one prediction that can be taken to data. Algorithmic homogeneity should amplify the transmission of U.S. monetary shocks to emerging-market capital flows, and it should do so only in stress regimes, once the amplification threshold ϕ^* has fallen below prevailing homogeneity (Proposition 3). We test that prediction on a panel of equity portfolio flows. The herding dimension of algorithmic behavior, which is our empirical counterpart of ϕ, does amplify capital outflows after U.S. monetary shocks, and the amplification appears only when global financial stress is high. The sign of the effect is robust. Its statistical significance is not, and it turns on two design choices that we would rather state here than have a reader find on their own.

7.1 Data, variables, and sample

The panel covers 19 emerging-market economies at quarterly frequency from 2000Q1 to 2024Q4. The dependent variable, Y_{it}, is the equity portfolio flow to country i in quarter t, drawn from the IMF Balance of Payments statistics (BPM6). The estimation sample contains $N = 1{,}636$ country–quarter observations.

The variable Y_{it} records how much equity investment foreign investors channelled into, or pulled out of, country i over the quarter. Y_{it} is built in two steps that replace the raw panel's annual-repeated series with genuine quarterly variation. Portfolio equity flows are downloaded from the IMF Balance of Payments statistics (BPM6) at quarterly frequency, on the liabilities side of the portfolio-investment-equity account (indicator `P_F5`, accounting entry `L_NIL_T`, the net incurrence of liabilities); `P_F5` (equity plus investment-fund shares) is preferred to the narrower `P_F51` because it maximizes historical coverage back to 2000Q1, and the two coincide almost exactly for economies where fund shares are a small part of the account. A positive value means nonresidents are increasing their equity claims on country i—a net portfolio inflow. The resulting series, in current USD, is divided by 10^9 so that Y_{it} is measured in USD billions. Two further points on how the coefficients read. The first is that Y_{it} is a flow and not a stock, so what we estimate is how a monetary shock moves the pace at which capital enters or leaves, not the level of holdings that have already accumulated. The second is that the BPM6 framework fixes both the sign convention and the boundary of what counts as a portfolio flow. Koepke & Paetzold (2024) document these conventions and argue that balance-of-payments portfolio-flow data are the right measure for the kind of macroeconomic and external-financing question we are asking, in contrast to the high-frequency fund-flow proxies such as EPFR, which cover only the mutual-fund and ETF subset of investors. The BPM6 series therefore lines up with the object the model actually speaks to, namely the total cross-border equity flow the sector intermediates, rather than a partial and selection-prone slice of it.

Algorithmic behavior is captured by *A_herding*, which is built from the BIS Locational Banking Statistics in three steps. First, for each country and quarter we compute the percentage change in total (bank and non-bank) external asset positions relative to the previous quarter. Second, we

average that same percentage change across every country in the panel in a given quarter to obtain a global benchmark, the common component of position growth that quarter. Third, $A_herding_{it}$ is the rolling Pearson correlation, over the trailing eight quarters (the current quarter and the seven before it, requiring at least six non-missing paired observations), between the country's own percentage change and the global benchmark. A country whose position changes closely follow the global benchmark over that two-year window has a high *A_herding*, meaning its intermediaries are, in the aggregate, behaving like everyone else; a country whose changes are not correlated with the benchmark has a low or negative value. The proxy is lagged for one quarter, $A_herding_{i,t-1}$, to avoid using information contemporaneously with r_t. *A_herding* is the empirical object the model identifies as the source of amplification.

The monetary shock r_t is the high-frequency U.S. monetary policy surprise from the San Francisco Fed's USMPD series, sign-flipped so that a positive value is an expansionary surprise. We want to be clear that this is not a proxy of convenience. It measures the same kind of statistical object the theory puts at the center of the mechanism. Recall that in the aggregation step of Section 3 the idiosyncratic components of algorithmic forecast errors wash out under the exact law of large numbers, while the common component survives and scales the variance of aggregate lending by ϕ. Homogeneity, in other words, works through the cross-sectional correlation of what funds do, not through the level or the speed of any single fund's activity. The finance literature measures herding in the same way, as the tendency of investors to move into and out of the same positions at the same time (Sias, 2004). That literature also draws the distinction we need, between spurious herding, where funds simply react alike to common fundamentals, and intentional herding, where the clustering comes from imitation itself; the distinction is set out in Bikhchandani & Sharma (2000) and has its theoretical roots in Banerjee (1992) and Scharfstein & Stein (1990). Because our specification interacts the proxy with an identified monetary shock and conditions on the stress regime and on country fundamentals, it filters out the mechanical comovement that common exposure alone would produce, and what remains is the correlated-behavior component that stands in for ϕ. The eight-quarter window is a compromise. It is long enough to estimate a correlation with usable precision and short enough to let the measure move with the state of the world, which Proposition 3 needs it to do, and the one-quarter lag keeps the synchrony we measure ahead of the flow response it is meant to explain. What the theory identifies as the source of amplification is this correlation, not the raw velocity of adjustment, which is why we build the proxy to capture synchrony rather than speed.

The stress regime is a dummy, $\text{Crisis}_t = \mathbb{I}(\text{VIX} > 30)$, equal to one when the VIX is above thirty. Every specification includes country fixed effects μ_i, and we cluster standard errors by country. The panel is mildly unbalanced. Sixteen of the nineteen economies are covered over the whole 2000Q1–2024Q4 window; China, Malaysia, and Nigeria enter later, once their external-position data become available, and Vietnam runs only through 2014. Table 3 gives the countries and their coverage.

Table 4 summarizes each variable, its definition, and its source.

Table 3: Emerging-market panel: countries and coverage

Country	ISO	Coverage	Quarters
Argentina	ARG	2000Q1–2024Q4	100
Brazil	BRA	2000Q1–2024Q4	100
Chile	CHL	2000Q1–2024Q4	100
China	CHN	2005Q1–2024Q4	80
Colombia	COL	2000Q1–2024Q4	100
Czech Republic	CZE	2000Q1–2024Q4	100
Hungary	HUN	2000Q1–2024Q4	100
India	IND	2000Q1–2024Q4	100
Indonesia	IDN	2000Q1–2024Q4	100
Malaysia	MYS	2002Q1–2024Q4	92
Mexico	MEX	2000Q1–2024Q4	100
Nigeria	NGA	2014Q1–2024Q4	44
Peru	PER	2000Q1–2024Q4	100
Poland	POL	2000Q1–2024Q4	100
Russia	RUS	2000Q1–2024Q4	100
South Africa	ZAF	2000Q1–2024Q4	100
Thailand	THA	2000Q1–2024Q4	100
Turkey	TUR	2000Q1–2024Q4	100
Vietnam	VNM	2005Q1–2014Q4	40
Total country–quarter observations			1,756

Notes. Coverage refers to quarters with a non-missing equity portfolio flow. The total of 1,756 country–quarter observations is the raw panel; the herding specification uses 1,636 observations because *A_herding* requires a rolling window and a one-quarter lag.

7.2 Econometric specification

We begin with a baseline panel specification that relates the equity portfolio flow to the U.S. monetary shock, to algorithmic behavior, and to their interaction:

$$Y_{it} = \alpha\, r_t + \beta\, A_{i,t-1} + \gamma\,(r_t \times A_{i,t-1}) + \mu_i + \varepsilon_{it}, \tag{26}$$

where $A_{i,t-1}$ is one of the two algorithmic proxies, lagged one quarter to mitigate simultaneity. Here α is the average flow response to a monetary surprise, β the level effect of algorithmic behavior, and γ, the coefficient we care about, tells us how far algorithmic behavior modulates the transmission of the shock. Country fixed effects μ_i absorb time-invariant heterogeneity across the nineteen economies, and we cluster the standard errors by country.

This average interaction, though, is not what the model predicts. Proposition 3 is a claim about state dependence: amplification should show up only when fundamentals are weak, which we proxy with the high-volatility regime $\text{Crisis}_t = \mathbb{I}(\text{VIX} > 30)$. To get at it we extend (26) to a triple interaction and keep every lower-order term, so that the triple coefficient is cleanly identified:

$$\begin{aligned} Y_{it} = {} & \alpha\, r_t + \beta\, A_{i,t-1} + \delta\, \text{Crisis}_t \\ & + \beta_4\,(A_{i,t-1} \times \text{Crisis}_t) + \beta_5\,(r_t \times A_{i,t-1}) + \beta_6\,(r_t \times \text{Crisis}_t) \\ & + \gamma\,(r_t \times A_{i,t-1} \times \text{Crisis}_t) + \mu_i + \varepsilon_{it}. \end{aligned} \tag{27}$$

Table 4: Variables, definitions, and sources

Variable	Definition	Source
Y_{it}	Quarterly equity portfolio flow to country i, the dependent variable (USD bn).	IMF Balance of Payments (BPM6).
r_t	High-frequency U.S. monetary policy surprise, sign-flipped so that a positive value denotes an expansionary surprise.	San Francisco Fed's USMPD series
A_herding	Rolling eight-quarter correlation of a country's position changes against a global benchmark; measures cross-country synchrony of behavior. Empirical analogue of ϕ. Entered with a one-period lag.	Constructed from BIS Locational Banking Statistics.
Crisis_t	Indicator equal to one when the VIX exceeds 30; defines the global financial-stress regime.	CBOE VIX.

Notes. Panel of 19 emerging-market economies, quarterly, 2000Q1–2024Q4. The estimation sample contains $N = 1{,}636$ country–quarter observations for the herding specification, fewer than the raw panel because *A_herding* requires a rolling window and a one-quarter lag.

With the three level terms (r_t, $A_{i,t-1}$, Crisis_t) and all three two-way interactions in the equation, γ measures the extra effect of the shock-behavior interaction that shows up specifically in the crisis regime, net of the average interaction (β_5), the average crisis effect on the shock (β_6), and the composition effect (β_4). Read this way, the two coefficients answer two different questions: β_5 is the interaction in normal times, and γ is how much that interaction changes once a crisis hits. A negative γ says that, under stress, more algorithmic homogeneity amplifies the capital outflow a contractionary U.S. shock sets off, which is what Proposition 3 predicts. Since $A_{i,t-1}$ enters both in levels and in every interaction, γ is identified from within-country variation in the joint occurrence of a monetary surprise, elevated homogeneity, and a stress regime, and not from cross-country differences in average levels.

7.3 Main results

Table 5 reports the estimates. Columns 1–3 set out the raw material of the mechanism before we condition on the state of the world. They ask whether a U.S. monetary surprise moves periphery equity flows at all, and whether the response is larger where intermediaries behave more synchronously. Columns 4–5 are where the model is actually tested. Proposition 3 does not say that homogeneity amplifies transmission on average; it says amplification switches on only once fundamentals weaken and the threshold ϕ^* falls below prevailing homogeneity. So the average interaction in columns 1–3 is the wrong place to look for it. What we want is the difference between tranquil and stress regimes, and that is exactly what the saturated triple interaction picks out. In practical terms this means following one coefficient across the table: the interaction of the shock with herding. If the theory holds, the piece of it that survives in tranquil times should fall toward zero, while the piece

that appears only in crises should carry the sign and the weight. Columns 4 and 5 differ only in whether we absorb push factors, which checks whether any state-dependent amplification we find really reflects algorithmic correlation and not the global-risk shock (the VIX) under another name. We read the estimates term by term below; the coefficient does move the way the theory says it should.

Table 5: Panel estimates of algorithmic amplification

	(1) Baseline	(2) Herding	(3) + Controls	(4) Triple	(5) Triple + push
Shock MP (r_t)	-2.965^{**}	-1.312	-3.089^{**}	-0.620	-0.439
	(1.121)	(0.995)	(1.252)	(1.299)	(0.763)
Crisis_t				-0.437^{**}	0.022
				(0.177)	(0.262)
$A_herding_{t-1}$		0.055	0.055	0.079	
		(0.322)	(0.322)	(0.349)	
$r_t \times A_herding_{t-1}$		-6.476^{**}	-6.914^{**}	0.140	
		(2.831)	(3.091)	(1.781)	
$r_t \times \text{Crisis}_t$				-0.634	-4.922^{**}
				(1.011)	(1.792)
$A_herding_{t-1} \times \text{Crisis}_t$				0.059	
				(0.357)	
$r_t \times A_herding_{t-1} \times \text{Crisis}_t$ (γ)				-9.581^{**}	-11.062^{**}
				(4.069)	(4.446)
GDP growth			0.005		-0.017
			(0.028)		(0.030)
Current account/GDP			-0.056		-0.059
			(0.035)		(0.036)
VIX					-0.019
					(0.013)
Country fixed effects	Yes	Yes	Yes	Yes	Yes
N	1,756	1,636	1,741	1,636	1,636
R^2	0.245	0.249	0.248	0.255	0.257

Notes. The dependent variable is the quarterly equity portfolio flow to country i (USD bn). Standard errors, clustered by country, are in parentheses. The monetary shock r_t is sign-flipped so that a positive value denotes an expansionary surprise; a negative coefficient therefore means a contractionary surprise reduces inflows. *A_herding* is the eight-quarter rolling correlation of a country's position changes with a global benchmark, lagged one quarter. $\text{Crisis}_t = \mathbb{I}(\text{VIX} > 30)$. Columns (1)–(3) report the baseline and its average interaction; columns (4)–(5) report the saturated triple interaction, in which γ tests the central hypothesis of state-dependent amplification. Macro controls are GDP growth and the current account/GDP; global push factors add the VIX. All columns include country fixed effects. $^{*}\,p < 0.1$, $^{**}\,p < 0.05$, $^{***}\,p < 0.01$.

The five columns map, term by term, onto the model. Columns 1–3 give us the two building blocks. The coefficient on the monetary shock is negative, significant, and stable around -3: a contractionary U.S. surprise pulls equity inflows out of the periphery, which is the basic transmission the center–periphery structure of Section 3 takes for granted. The average interaction $r_t \times A_herding$ is more telling. It is negative and significant, between -6.476 and -6.914, so countries whose intermediaries move more in step with the global benchmark take larger outflows from the same shock. That is the unconditional version of Proposition 1, correlated behavior amplifying transmission, and the sign is the one the model predicts. On its own, the level of *A_herding* is insignificant, again as predicted: what moves flows is not the presence of homogeneity but its interaction with the shock it amplifies.

Columns 4–5 carry the paper's central result. Proposition 3 predicts that amplification is not a constant but a state-dependent phenomenon, appearing only once fundamentals weaken and the threshold ϕ^* drops below prevailing homogeneity, and the saturated triple interaction bears this out. The two-way interaction $r_t \times A_herding$ collapses to an insignificant 0.140, so in tranquil times herding does not amplify. The triple interaction with the crisis regime, by contrast, is large, negative, and significant: $\gamma = -9.581$, and it rises to -11.062 once we absorb global push factors. The amplification thus shows up entirely in the stress regime, which is the empirical face of the sign reversal at the heart of the model. It matters here that γ gets stronger, not weaker, when push factors enter, with the direct global-risk channel loading onto $r_t \times$ Crisis at -4.922. That tells us the herding amplification is its own conditional effect, and not the global-risk shock wearing a different label.

The magnitude is economically meaningful as well. Taking a country from the median to a high level of algorithmic homogeneity during a stress regime multiplies the sensitivity of its equity flows to a U.S. monetary surprise by a sizable factor, which sits comfortably with the amplification factor $\mathcal{A} = \sqrt{\phi_H/\phi_L}$ that our calibration puts near two at the benchmark. So the estimate has the right sign and is of roughly the order of magnitude the theory implies, which is what ties the empirical γ back to the analytical object in Proposition 1.

There is one gap in this link that we would rather flag than paper over. Proposition 1 is a statement about the variance of the aggregate lending response: homogeneity leaves the mean unchanged and raises the dispersion. Our coefficient γ, on the other hand, picks up a shift in the conditional mean of flows. The two are not in conflict. When the shock carries a contractionary sign, a state-dependent rise in the second moment will indeed surface as a mean outflow in the crisis regime. But they are not the same object either, and a test that did full justice to Proposition 1 would go after the conditional variance of flows directly, for example with a regime-conditional heteroskedasticity specification that lets the herding proxy enter the variance equation. We therefore read the conditional-mean evidence as consistent with the proposition without standing in for a proper second-moment test. That test is the extension we would put first, all the more so because the work closest to ours in structure (Meng & Chen, 2026; Qiu & Han, 2026) goes at tail risk directly.

It is worth being precise about which feature of algorithmic behavior does the work here. The model points to the correlation of forecast errors across funds, not to the raw speed at which any one fund adjusts its position. Two intermediaries can react to a shock at exactly the same speed and still, if their models are uncorrelated, have their errors cancel in the aggregate, leaving no amplification behind. Only when their behavior is synchronized, at high ϕ, does the common component survive aggregation and move flows. That is why we build the proxy around synchrony (*A_herding*), which the theory names as the source of amplification, rather than around velocity, which the model gives us no reason to expect to matter. The distinction is more than semantic. It is what sets our mechanism apart from accounts that stress the mechanical speed of algorithmic or high-frequency trading, and it is why a policy of slowing markets down would miss the margin our results point to.

This refines a result already in the transmission literature. Kalemli-Özcan (2019) shows that U.S. policy spills over to emerging markets more strongly than to advanced ones, working through global risk perceptions, and our advanced-economy falsification reproduces that asymmetry. We take it a step further by pinning down which feature of the intermediation sector carries the amplification, behavioral correlation and not speed, and by showing that it works only in the high-VIX regime where those risk perceptions dominate. That literature tends to treat the intermediary sector as a conduit for a risk-premium shock; our evidence instead places the amplification in the cross-sectional correlation of the intermediaries themselves, which is what ϕ stands for.

7.4 Robustness

We now put the central estimate through a battery of checks. The estimate is the negative triple-interaction coefficient γ on $r_t \times A_herding \times \text{Crisis}$, which measures the extra emerging-market equity outflow per unit of U.S. monetary shock that synchronized algorithmic behavior produces under stress. Every check moves away from the main specification, column 4 of Table 5, along exactly one dimension, holding fixed the country fixed effects, the high-frequency USMPD shock, the full lower-order interaction tree, equity portfolio flows (BPM6) in USD billions, the VIX-based crisis regime, and the country-clustered standard errors. That baseline reproduces the reported coefficient of -9.581 (SE 4.069, $p = 0.030$) on $N = 1{,}636$, and Table 6 collects the twelve specifications.

The result holds up in sign and in magnitude, and it is significant under the main specification and under the inference schemes best suited to the setting. The coefficient γ is negative in all twelve specifications, significant at the 10% level in nine and at the 5% level in seven, including under the time-clustering and heteroskedasticity-robust schemes that fit a panel of capital flows responding to a common U.S. shock. The point estimate is stable across the inference, outlier, and sample-composition checks. Significance lapses in two cases only, and we would rather name them than tuck them away: dropping the 2008–09 crisis, which takes the most informative stress episode out of a mechanism that is state-dependent by design, and redefining the crisis regime by calendar years instead of by the VIX. That the result is sensitive to how we operationalize "crisis" tells us something about the evidence; it is not a weakness we want to hide.

Table 6: Robustness battery for the herding-amplification coefficient γ

Specification	γ (triple)	Std. err.	N
Baseline (VIX>30, cluster country)	-9.581^{**}	(4.069)	1,636
With push factors	-8.831^{*}	(4.487)	1,636
Two-way FE (country + year)	-6.309^{*}	(3.266)	1,636
Cluster SE by time	-9.581^{***}	(2.975)	1,636
Heteroskedasticity-robust SE	-9.581^{**}	(4.814)	1,636
Winsorize Y (1/99 pct)	-7.716^{**}	(2.770)	1,636
Dynamic panel (lagged dep.)	-8.850^{**}	(3.807)	1,632
Exclude China	-8.452^{**}	(3.943)	1,556
Exclude 2008–09 GFC	-11.232	(6.549)	1,492
Exclude 2020 COVID	-7.227^{**}	(2.771)	1,564
Alt. regime: crisis years	-8.239	(6.158)	1,636
Alt. shock: JK pure-monetary	-11.397	(9.162)	1,636

Notes. Each row re-estimates γ under the stated departure from the baseline specification of column (4) of Table 5. Dependent variable: equity portfolio flow (BPM6), USD billions. Standard errors in parentheses, clustered by country except where noted. "Alt. shock: JK pure-monetary" uses the Jarociński & Karadi (2020) pure-monetary surprise, which purges the central-bank-information component. All estimates are generated from the panel of 19 emerging markets, 2000Q1–2024Q4, via a within (country fixed-effects) estimator following the triple-interaction specification (27). $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.

The checks fall into four groups. Appendix B sets out the purpose, procedure, and verdict for each one; here we summarize what they show.

The first group concerns inference. Adding year fixed effects on top of country effects leaves $\gamma = -6.309$, still significant at the 10% level, and this is a demanding test because year effects soak up part of the common stress variation the interaction relies on. Clustering the standard errors by time rather than by country, which is the scheme that matches the cross-country comovement the mechanism implies, gives back the same point estimate with a smaller standard error, significant at the 1% level. Heteroskedasticity-robust errors leave the coefficient where it was, significant at 5%. Under the schemes best suited to the setting, in short, the result is at its strongest.

The second group concerns influential observations. Winsorizing the dependent variable at the 1st and 99th percentiles gives $\gamma = -7.716$, significant at 5%, so the amplification lives in the body of the data rather than in a handful of extreme quarters. A dynamic panel with a lagged dependent variable gives $\gamma = -8.850$, again at 5%, so persistence in flows is not what is driving the result.

The third group varies the sample. Dropping China, the largest recipient, leaves $\gamma = -8.452$ at 5%, so this is not a China artifact. Dropping the March-2020 COVID quarter, the sharpest emerging-market outflow on record, leaves $\gamma = -7.227$, also at 5%, so the pandemic is not carrying the result either. Dropping the 2008–09 global financial crisis is a different story. The coefficient stays negative and in fact grows in magnitude, to -11.232, but significance slips to $p = 0.103$. This is the price we expect to pay for removing the single most informative stress episode from a state-dependent mechanism: the sign and the point estimate survive, the precision does not. We read it as consistent with the mechanism rather than against it.

The fourth group probes the design choices, and this is where the two qualifications sit. Redefining the crisis regime by calendar years (2008–09 and 2020–21) instead of by the VIX leaves γ negative and

of similar magnitude, −8.239, but insignificant ($p = 0.198$). This is the most consequential caveat in the battery. The VIX-based regime tracks market stress continuously and delivers a significant estimate; the coarser calendar-year regime does not. Our reading is that the VIX definition is the better counterpart of the model's stress state, since Proposition 3 describes a continuously falling threshold rather than a step function tied to the calendar. Still, a reader deserves to know that the headline significance turns on this choice, and Table 6 puts it in plain view. Replacing the USMPD surprise with the Jarociński & Karadi (2020) pure-monetary shock leaves γ negative and larger, −11.397, but with a wider standard error and no conventional significance. The sign and the economic message survive the change of shock series; the precision does not.[1]

One further test deserves a place here, because it speaks to the gap between what Proposition 1 states and what γ measures. The proposition is about the *variance* of the aggregate response, while the triple interaction identifies a shift in the conditional mean. The natural way to confront the proposition on its own terms is to let herding enter a variance equation directly. We do this in two steps: we take the residuals from the mean equation of column (4), and we regress their log squared value on *A_herding*, the crisis dummy, and their interaction. If homogeneity raises the dispersion of flows in stress, and does so more than in tranquil times, the interaction term should be positive. It is: the coefficient on *A_herding* × Crisis is +0.819, with the expected sign. But the standard error is 0.780 and the estimate is not significant at conventional levels ($p = 0.294$). We read this as suggestive rather than confirmatory. The second moment moves the way the theory says it should, but a panel of this length does not have the power to pin it down, and we would not want to lean on it. A properly powered second-moment test—one that models the conditional variance of flows directly rather than backing it out of mean-equation residuals—is in our view the first extension this design calls for, and we say so plainly rather than claim more than the data will bear.

To sum up, the evidence lines up with the model's central prediction. The herding dimension of algorithmic behavior amplifies the international transmission of U.S. monetary shocks to emerging markets, and it does so only under stress, as Proposition 3 requires, with the correlation of algorithmic behavior being the margin that carries the effect, much as Proposition 1 would lead one to expect. The sign and the state-dependence hold throughout; the statistical significance holds under the baseline and the appropriate inference schemes, subject to the two qualifications on regime definition and crisis-sample coverage that we have reported in full.

[1]Two further sensitivities are worth flagging in the same spirit, since a careful reader will want them. First, the VIX threshold itself. The estimate is significant at the 30 cutoff of the baseline and at the 90th percentile of the VIX (≈ 28), but a higher 35 cutoff leaves so few stress quarters that the coefficient loses precision and even flips sign, while a lower 25 cutoff dilutes the regime and weakens it ($p = 0.22$). The effect lives in a band around VIX $\approx$ 28–30, which is where the model's stress state is most cleanly identified, but we do not hide that it is a band. Second, temporal stability. Splitting the sample at 2012 leaves the coefficient negative in both halves, significant in 2000–2011 ($p = 0.04$) and larger but imprecise in 2012–2024 ($p = 0.21$); the later half excludes the 2008–09 episode, so the loss of precision is the same identification point made in the sample-composition checks above, not a separate failure. Neither sensitivity changes the sign or the economic reading; both are reasons we frame the significance as conditional rather than settled.

8 Policy Implications and Conclusion

The paper began with a puzzle about the role of AI in financial trading: why does the same rise of algorithmic intermediation look stabilizing in some accounts and destabilizing in others? Our answer is that the margin that matters is not the presence of algorithms but their homogeneity. We built a two-region DSGE model in which a single parameter ϕ measures the correlation of forecast errors across algorithmic funds, and it delivers three results: homogeneity amplifies transmission through the variance of aggregate lending (Proposition 1), diversity attenuates it (Proposition 2), and the threshold separating the two regimes falls as fundamentals deteriorate (Proposition 3). The panel evidence of Section 7 confirms the central prediction: the herding dimension of algorithmic behavior amplifies emerging-market capital outflows in response to U.S. monetary shocks, and does so only in stress regimes. Four implications for policy follow, each from a specific proposition, and several cut against the grain of current macroprudential practice.

The first concerns the object of regulation. The prevailing macroprudential frame, inherited from the banking literature, targets the size and concentration of institutions, but Proposition 2 implies that this frame is miscalibrated for algorithmic intermediation. A large non-bank sector is not dangerous per se; a homogeneous one is. A regulator that caps the size of the algorithmic sector while leaving its models correlated addresses the wrong margin, and may even raise fragility if the cap forces consolidation onto a common surviving platform. The appropriate instrument is a measure of model diversity—an index of the correlation of positioning or of forecast errors across funds—which has no counterpart in size-based capital regulation and would need to be constructed anew. The logic parallels a market-design argument made in a different context: Budish et al. (2015) show that the high-frequency trading arms race is a symptom of a flawed continuous-market design and propose frequent batch auctions to neutralize the value of correlated speed. Their target is the mechanism that makes synchronized speed profitable; ours is the model correlation that makes synchronized trading destabilizing. In both cases the effective lever is structural—the design of the market or the diversity of its participants—rather than a restriction on volume.

We should be candid about what makes this instrument hard to wield, since naming the target is easier than hitting it. Model diversity has the character of a public good. Each fund has a private incentive to adopt whatever model performs best, and when one architecture pulls ahead the individually rational response is to converge on it, which is precisely the behavior that erodes the diversity the system relies on. No single fund internalizes the systemic cost of that convergence, so diversity will be underprovided by the market and a regulator who merely exhorts firms to differ will accomplish little. There is also a genuine trade-off to acknowledge rather than assume away. Convergence on a strong common model is not pure loss: it can improve the average quality of pricing in normal times, so a mandate for diversity trades some efficiency in tranquil periods for resilience in stressed ones. Our results do not tell a regulator where to strike that balance; they establish only that the balance exists and that the current frame, fixated on size, does not see it. What they do suggest is that the natural instrument is one supervisors already possess in adjacent

form—model validation and model-risk review, currently run fund by fund for prudential soundness, could be repurposed to measure correlation *across* funds—so the practical step is less the invention of a new tool than the redirection of an existing one toward a systemic object.

A second implication runs directly counter to a natural regulatory instinct: to publish a standard risk model, mandate a common stress scenario, or require firms to benchmark against a supervisory model. Proposition 1 warns that this instinct is self-defeating, because any measure that pushes funds toward a common model raises ϕ and therefore raises the amplification factor $\sqrt{\phi_H/\phi_L}$. Standardization that improves the transparency of any single institution can degrade the stability of the system, since it manufactures precisely the correlated errors the model identifies as the source of herding. Regulators should therefore design disclosure requirements to reveal whether funds are converging on common models, not to impose one.

The third implication is about timing. Because Proposition 3 makes the amplification threshold ϕ^* state-dependent, a level of homogeneity that sits comfortably below the threshold in tranquil times can exceed it in a crisis without ϕ itself changing. A supervisor who measures homogeneity in calm periods and compares it to a fixed benchmark will systematically understate the risk, because the benchmark against which a supervisor should judge ϕ falls exactly when fundamentals deteriorate. A supervisor must evaluate homogeneity metrics against a state-contingent threshold that tightens as local fundamentals weaken, an approach closer to a stress test on model diversity than to a static concentration limit.

The sharpest implication is dynamic and lies beyond the static model. If funds imitate strategies that performed well, ϕ rises endogenously during calm periods, precisely when ϕ^* is high and the extra homogeneity appears harmless. Diversity is thus consumed as a common resource in good times, sowing the fragility that the falling threshold then exposes in the next crisis. This reframes the policy problem as counter-cyclical management of diversity: the time to encourage heterogeneous models, alternative data, and divergent architectures is when markets are calm and the incentive to imitate is strongest, not after correlated positioning has already built up. Emerging-market authorities, who receive rather than generate the global cycle, have a particular interest in attracting a diverse rather than a monolithic set of algorithmic investors, since it is diversity, not the sheer volume of inflows, that determines whether the non-bank sector stabilizes or amplifies the flows they cannot control.

The broader message is that the rise of algorithmic and AI-driven intermediation need not make the international financial system more fragile. Its effect is conditional: the same technology stabilizes capital flows when models are diverse and amplifies them when models converge. The policy task is not to resist algorithmic intermediation but to preserve its diversity—a variable that current regulation does not measure and that erodes fastest when it is least missed. Making model diversity observable, and defending it counter-cyclically, is the first-order priority that the model implies.

A Derivations

This appendix derives every optimality condition of the model step by step. Where a standard second-order simplification is used, we state it explicitly rather than absorb it silently.

A.1 The household problem

The household maximizes (1) subject to the budget constraint (2) and the capital law of motion $K_t = (1-\delta)K_{t-1} + I_t$. Let λ_t be the multiplier on the budget constraint and $q_t\lambda_t$ the multiplier on the capital law of motion, so that q_t is Tobin's marginal q. The Lagrangian is

$$\begin{aligned}\mathcal{L} = \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \Bigg\{ & \frac{C_t^{1-\sigma}}{1-\sigma} - \chi \frac{N_t^{1+\nu}}{1+\nu} + q_t\lambda_t\big[(1-\delta)K_{t-1} + I_t - K_t\big] \\ & + \lambda_t \Big[W_t N_t + R_{t-1}D_{t-1} + R_t^K K_{t-1} + \Pi_t - C_t - D_t - I_t \\ & \quad - \tfrac{\psi_I}{2}\big(\tfrac{I_t}{K_{t-1}} - \delta\big)^2 K_{t-1} \Big] \Bigg\}.\end{aligned}$$

Consumption and deposits. The first-order condition with respect to C_t is $C_t^{-\sigma} = \lambda_t$, so the multiplier equals marginal utility. The condition with respect to D_t is $-\lambda_t + \beta\,\mathbb{E}_t[\lambda_{t+1}R_t] = 0$. Substituting $\lambda_t = C_t^{-\sigma}$ and $\lambda_{t+1} = C_{t+1}^{-\sigma}$ gives the consumption Euler equation (3):

$$1 = \beta\,\mathbb{E}_t\left[\left(\frac{C_{t+1}}{C_t}\right)^{-\sigma} R_t\right].$$

Labor. The first-order condition with respect to N_t is $-\chi N_t^{\nu} + \lambda_t W_t = 0$. Substituting $\lambda_t = C_t^{-\sigma}$ and rearranging yields the labor-supply schedule (4):

$$\chi N_t^{\nu} C_t^{\sigma} = W_t.$$

Investment. Differentiating with respect to I_t, and noting that $\partial\big[\tfrac{\psi_I}{2}(\tfrac{I_t}{K_{t-1}} - \delta)^2 K_{t-1}\big]/\partial I_t = \psi_I\big(\tfrac{I_t}{K_{t-1}} - \delta\big)$, the condition is

$$-\lambda_t\Big[1 + \psi_I\big(\tfrac{I_t}{K_{t-1}} - \delta\big)\Big] + q_t\lambda_t = 0 \quad\Longrightarrow\quad q_t = 1 + \psi_I\Big(\frac{I_t}{K_{t-1}} - \delta\Big).$$

At the steady state $I/K = \delta$, so $q = 1$, as required. Differentiating with respect to K_t gives

$$q_t\lambda_t = \beta\,\mathbb{E}_t\Big[\lambda_{t+1}\big(R_{t+1}^K - \tfrac{\partial \mathrm{AC}_{t+1}}{\partial K_t}\big) + q_{t+1}\lambda_{t+1}(1-\delta)\Big],$$

where AC_{t+1} is the adjustment cost dated $t+1$. The term $\partial \mathrm{AC}_{t+1}/\partial K_t = \tfrac{\psi_I}{2}\big(\delta^2 - (I_{t+1}/K_t)^2\big)$ vanishes to first order around the steady state (where $I/K = \delta$), which is the standard simplification.

Dropping it and using $\lambda_{t+1}/\lambda_t = (C_{t+1}/C_t)^{-\sigma}$ delivers the investment Euler equation (5):

$$q_t = \beta\, \mathbb{E}_t \left[\left(\frac{C_{t+1}}{C_t} \right)^{-\sigma} \left(R^K_{t+1} + q_{t+1}(1-\delta) \right) \right].$$

A.2 The firm problem

The firm maximizes profit $Y_t - W_t N_t - R^K_t K_{t-1}$ subject to the technology (6). The first-order conditions equate marginal products to factor prices:

$$\frac{\partial Y_t}{\partial N_t} = (1-\alpha)\frac{Y_t}{N_t} = W_t, \qquad \frac{\partial Y_t}{\partial K_{t-1}} = \alpha \frac{Y_t}{K_{t-1}} = R^K_t,$$

which are the factor prices stated in the text. The loan rate faced by firms is $\tilde{R}^K_t = R_t + \mu_t$ with the premium μ_t given by (7).

A.3 Traditional funds

Traditional funds maximize (8). Differentiating the objective with respect to L^T_t,

$$(\tilde{R}^K_t - R_t) - \psi_T \Big(\frac{L^T_t}{L^T_{t-1}} - 1 \Big) = 0.$$

Using $\tilde{R}^K_t - R_t = \mu_t$ gives the sluggish lending rule (9), $\mu_t = \psi_T \big(L^T_t / L^T_{t-1} - 1 \big)$: lending adjusts only gradually, at a pace set by ψ_T.

A.4 Algorithmic funds and aggregation

Each fund j forms the minimum-MSE forecast (13) and lends according to the static rule (16),

$$L^A_{j,t} = \bar{L}_A - \gamma_1 \hat{\mu}_t + \gamma_2 \big(\theta^{MP} a\, s^{MP}_{j,t} + \theta^{LOC} b\, s^{LOC}_{j,t} \big).$$

Substituting the signals (10)–(11) with the noise decomposition (12) and integrating over $j \in [0,1]$, the exact law of large numbers for a continuum (Sun, 2006) sets $\int_0^1 u^{MP}_{j,t}\, dj = \int_0^1 u^{LOC}_{j,t}\, dj = 0$, because the idiosyncratic components are mean-zero and independent across funds. The common components $\zeta^{MP}_t, \zeta^{LOC}_t$ do not depend on j and pass through the integral unchanged. The aggregate lending rule is therefore (17):

$$L^A_t = \bar{L}_A - \gamma_1 \hat{\mu}_t + \gamma_2 \Big[\theta^{MP} a \big(MP_t + \sqrt{\phi}\, \zeta^{MP}_t \big) + \theta^{LOC} b \big(F_t + \sqrt{\phi}\, \zeta^{LOC}_t \big) \Big].$$

Two properties follow directly. When $\phi = 0$ the common terms vanish and the aggregate forecast is exact. The sensitivity of aggregate lending to the common noise is $\partial L^A_t / \partial \zeta^{MP}_t = \gamma_2 \theta^{MP} a \sqrt{\phi}$, so the variance of the aggregate response to a monetary shock is proportional to ϕ—the content of Propositions 1 and 2.

A.5 Market clearing

Total lending is $L_t = L_t^T + L_t^A$, with households allocating the fixed share ω of deposits to traditional funds. The goods market clears, $Y_t = C_t + I_t$, and the U.S. policy rate follows the Taylor rule (18). Together with the household and firm conditions above, these close the model.

B Detail of the Robustness Battery

This appendix documents the purpose, procedure, and verdict of each check in the robustness battery of Section 7.4 and Table 6, in the order in which the checks are grouped. All estimates start from the main-text specification of column (4) of Table 5 and depart from it in exactly the one dimension named.

B.1 Inference

Two-way (country and year) fixed effects. Country fixed effects alone leave common global shocks in the error term; adding year fixed effects absorbs any disturbance that hits all countries in a given year, so identification comes only from differences across countries within the same year. The check is demanding here because the crisis regime is itself largely a common time series, so year effects absorb part of the variation the mechanism operates through. Result: $\gamma = -6.309$ (SE 3.266, $p = 0.069$). The coefficient stays negative and significant at the 10% level even under this demanding control. Robust.

Clustering by time rather than by country. Country clustering allows arbitrary correlation of errors within a country over time; time clustering allows arbitrary correlation across countries within the same quarter—the natural concern when a global shock moves all peripheries together. Result: $\gamma = -9.581$ (SE 2.975, $p = 0.002$). The same point estimate becomes significant at the 1% level. Under the inference scheme that matches the cross-country comovement the mechanism predicts, the result is at its strongest.

Heteroskedasticity-robust standard errors. White/HC errors relax the constant-variance assumption, essential when capital flows in levels across countries of very different size have residual variance that scales with the country. Result: $\gamma = -9.581$ (SE 4.814, $p = 0.047$). Unchanged and significant at 5%. Robust.

B.2 Influential observations

Winsorizing the dependent variable at the 1st and 99th percentiles. Capital-flow data contain extreme quarterly values—sudden stops, surges—that can dominate a levels regression. Winsorizing caps the most extreme 1% in each tail while retaining every observation. Result: $\gamma = -7.716$ (SE 2.770, $p = 0.012$). The amplification is a feature of the bulk of the data, not of a few extreme episodes. Robust.

Dynamic panel with a lagged dependent variable. Capital flows are persistent; omitting that persistence can bias the coefficients. With T large relative to the Nickell bias, the lagged dependent variable serves as a stability check. Result: $\gamma = -8.850$ (SE 3.807, $p = 0.032$). Persistence in flows does not account for the amplification. Robust.

B.3 Sample composition

Excluding China. China is the largest recipient and could dominate a levels regression by scale. Result: $\gamma = -8.452$ (SE 3.943, $p = 0.047$). Not a China artifact. Robust.

Excluding the 2008–09 global financial crisis. The 2008–09 crisis is the most extreme stress episode and might single-handedly generate a state-dependent result; dropping those years is a demanding check for a mechanism active only under stress. Result: $\gamma = -11.232$ (SE 6.549, $p = 0.103$). The coefficient remains negative and grows in magnitude, but significance lapses to the 10.3% level. Removing the most informative stress episode from a state-dependent mechanism reduces precision—exactly what one expects—while leaving sign and point estimate intact. Consistent with, not contrary to, the mechanism.

Excluding the 2020 COVID quarter. The March-2020 reversal was the sharpest emerging-market outflow on record. Result: $\gamma = -7.227$ (SE 2.771, $p = 0.018$). The result does not hinge on the COVID episode. Robust.

B.4 Regime and shock definition

Redefining the crisis regime by calendar years. The baseline defines the crisis regime by VIX > 30; an alternative defines it by calendar years (2008–09 and 2020–21). Result: $\gamma = -8.239$ (SE 6.158, $p = 0.198$). The coefficient remains negative and of similar magnitude, but significance lapses. This is the single most important qualification in the battery: the VIX-based regime, which tracks market stress continuously, delivers a significant estimate, whereas the coarser calendar-year regime does not. We regard the VIX-based definition as the more faithful counterpart of the model's stress state, but report the sensitivity openly, because a reader is entitled to know that the headline significance depends on this choice.

Alternative shock: the Jarociński–Karadi pure-monetary series. The identified monetary innovation can be measured by the high-frequency USMPD surprise (baseline) or by the Jarociński & Karadi (2020) pure-monetary shock, which purges the central-bank-information component. Result: $\gamma = -11.397$ (SE 9.162, $p = 0.230$). The sign and economic message survive the change of shock series; the precision does not. Consistent with the mechanism, weaker under this alternative.

B.5 Reproducibility

All estimates are generated from the panel of 19 emerging markets and 11 advanced economies, 2000Q1–2024Q4, via a within (country fixed-effects) estimator following the triple-interaction specification (27) with the high-frequency USMPD shock and the VIX-based crisis regime. The baseline reproduces the main-text coefficient (-9.581, SE 4.069, $N = 1{,}636$); each row of Table 6 departs from that baseline in exactly the one dimension named.

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